

Comprehensive Analysis of Total Sediment Load Quantification in Fluvial Systems

Executive Summary

Quantifying total sediment transport is a fundamental requirement in river engineering and fluvial geomorphology. The total sediment load (q_s) is defined as:

$$q_s = q_b + q_{ss}$$

where:

- q_b : bed load
- q_{ss} : suspended load

A critical distinction must be made between:

- **Bed-material load**
 - Derived from the stream bed
 - Controlled by local hydraulics
- **Wash load**
 - Fine particles (silt and clay)
 - Controlled by watershed supply processes rather than local flow conditions

Two Primary Methodologies

1. **Divided Approach (Mechanistic)**
 - Separates bed load and suspended load
 - Example: Einstein method
 2. **Total-Load (Lumped) Approach (Empirical)**
 - Predicts total transport directly
 - Based on variables such as velocity (U), slope (S), and grain size (d)
-

Key Insights

- Empirical formulas may differ by a factor of **2–5**
 - Method selection depends on:
 - Sediment size
 - Transport regime (Rouse number)
 - Available hydraulic data
 - **Einstein method** → most physically rigorous
 - **Ackers–White and Yang** → preferred for engineering practice
-

1. Fundamentals of Total Sediment Load

1.1 Core Definitions

Sediment transport consists of two modes:

- **Bed Load** (q_b)
 - Moves near the bed
 - **Suspended Load** (q_{ss})
 - Distributed throughout the flow depth
 - Maintained by turbulence
-

1.2 Bed-Material Load vs. Wash Load

Load Type	Source	Characteristics	Control Mechanism
Bed-Material Load	River bed	Bed load + suspended bed material	Local hydraulics
Wash Load	Watershed	Fine particles (silt/clay), not in bed	Supply-controlled

1.3 Transport Regimes (Rouse Number)

$$P = \frac{V_s}{\kappa u_*}$$

Bed-load dominant: $P > 2.5$

- **Mixed load:** $1.2 < P < 2.5$
- **Suspension dominant:** $P < 1.2$

2. Approaches to Total Load Estimation

2.1 Methodological Frameworks

(1) Divided Approach (Microscopic / Analytical)

- Treats bed load and suspended load separately
- Physics-based
- Example: Einstein method

(2) Lumped Approach (Macroscopic / Empirical)

- Uses dimensional analysis and data fitting
- Relates transport to:

$$q_s = f(U, S, d)$$

2.2 Stream Power Concept

$$\omega = \tau U$$

- Introduced by **Bagnold (1966)**
 - Sediment transport is driven by **energy availability**
 - Foundation of many total-load formulas
-

3. Major Total-Load Formulas

3.1 Summary of Methods

Method	Coefficient Type	Sensitivity	Key Notes
Engelund–Hansen	Single constant (0.1)	Low	Simple; uses total shear stress; calibrated for sand-bed rivers
Yang	Variable coefficients	High	Physically insightful; sensitive to settling velocity and critical velocity
Molinas–Wu	Fixed constants	Moderate	Stable; no threshold term
Ackers–White	Multi-parameter	High	Complex; uses dimensionless grain size D_*

3.2 Key Characteristics

- **Engelund–Hansen**
 - Includes form drag implicitly
 - Often overpredicts
 - Suitable for quick estimates
 - **Yang**
 - Based on unit stream power (VS); outputs total sediment concentration (C_t , ppm) rather than volumetric transport rate
 - Requires critical velocity U_{cr}
 - **Molinas–Wu**
 - Smooth across regimes
 - Numerically stable
 - **Ackers–White**
 - Includes threshold behavior
 - Performs well for mixed sediment
-

4. Performance Evaluation

4.1 Comparative Example

Given:

- $H = 3 \text{ m}, S = 0.0003, D_{50} = 0.6 \text{ mm}$

Method	$q_t(\text{m}^2/\text{s})$	Ratio
Einstein (reference)	1.194×10^{-4}	1.00
Ackers–White	1.12×10^{-4}	0.94
Molinas–Wu	1.65×10^{-4}	1.38
Yang	2.26×10^{-4}	1.89
Engelund–Hansen	4.69×10^{-4}	3.93

Conclusion

- Ackers–White matches best
 - Engelund–Hansen overpredicts significantly
-

4.2 Reasons for Differences

- **Driving Stress**
 - Einstein → grain shear stress
 - Lumped → total shear stress
 - **Calibration**
 - Different datasets and environments
 - **Modeling Philosophy**
 - Physics-based vs empirical
-

5. Decision Framework

Step 1: Sediment Size

- Silt/clay → Wash load (not modeled)
 - Sand → Proceed
 - Gravel → Bed-load formulas
-

Step 2: Transport Regime

- $P < 1.2$ → Yang, Engelund–Hansen, Molinas–Wu
 - $1.2 < P < 2.5$ → Ackers–White, Molinas–Wu
 - $P > 2.5$ → Bed-load formulas or Ackers–White
-

Step 3: Data & Purpose

- Limited data → Engelund–Hansen
 - Velocity known → Yang, Molinas–Wu
 - Engineering → Yang, Ackers–White
 - Research → Einstein
-

6. Advanced Topics

6.1 Sediment Mixtures

$$\theta_{ci} = \xi_i \theta_c$$

- **Hiding effect:** small grains shielded
 - **Exposure effect:** large grains more exposed
-

6.2 Measurement Challenges

- **Suspended load**
 - Depth-integrating samplers
 - Point samplers
 - **Bed load**
 - Helley–Smith samplers
 - Highly variable and difficult to measure
-

6.3 Critical Engineering Warning

⚠ Maintain consistency in shear stress:

- Einstein → **grain shear stress**
- Engelund–Hansen → **total shear stress**

Mixing them can cause **major errors**

Final Takeaway

- No single formula is universally correct
- Always:
 - Understand assumptions
 - Match method to conditions
 - Compare multiple methods